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Software Solutions & Digital Services
Building Innovation. Delivering Excellence.

FINAL PROJECT

DevFlow CRM

CRM Built Specifically for Software Houses

Full-Stack Development Internship — U Devs 2026

Project	DevFlow CRM — Software House CRM
Project Type	Final Full-Stack Internship Project
Technology	PERN + JWT + Redux Toolkit + Bootstrap/Tailwind + Material Icons
Start Date	25 August 2026
End Date	02 September 2026
Organization	U Devs SMC PVT Limited
Powered By	U Devs SMC PVT LTD — 2026
Project Owner	U Devs Software & Digital Solutions

Core Product Goal: Build a production-style CRM that models how a software house manages leads, clients, projects, requirements, tasks, documents, invoices, support, teams and internal operations from one role-based dashboard.

Implementation standard: Responsive UI, reusable components, secure REST APIs, relational database design, role-based access control, validation, error handling, auditability and a clean Git-based development workflow.

Prepared / Issued By

Usama Aslam — Founder & CEO, U Devs

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1. Project Overview & Table of Contents

No.	Section	Purpose
01	Project Overview	Scope, objectives and success definition
02	Technology & Architecture	PERN, JWT, Redux, API and UI standards
03	Roles & Permissions	Role-based dashboard and access matrix
04	Authentication & Security	Login, registration, verification, reset, 2FA
05	CRM Core	Leads, prospects, clients, contacts, companies, deals
06	Project Management	Projects, milestones, tasks, assignments and deadlines
07	Client Portal	Client login, project visibility, tickets, messages, documents
08	Requirement Vault	Versioned requirements, attachments, comments and approvals
09	Dashboard & Analytics	KPIs, charts, pipeline, workload and activity
10	Frontend Screens	Page/component inventory and navigation flow
11	Redux & API Layer	Store slices, async actions, Axios/API organization
12	Backend & Database	Express structure, controllers, services and schema
13	Workflows	End-to-end software house CRM processes
14	Testing & Security	Validation, RBAC, API testing and acceptance
15	Timeline & Deliverables	25 Aug to 02 Sep implementation plan
16	Bonus Features	Advanced additions for extra credit

Executive Brief

DevFlow CRM is a vertical CRM for software houses rather than a generic sales-only CRM. The application should connect business development with delivery: a lead becomes a prospect, then a client, then a project; project requirements become milestones and tasks; approved work is traceable through documents, invoices, support tickets and client communication.

Recommended product principle: every important business action should leave a visible record — owner, date/time, status, comments and audit history. This makes the project feel like a real internal platform instead of a collection of CRUD screens.

Primary success condition: a user can authenticate, reach the correct role-based dashboard, create and update CRM records, move a deal through the pipeline, convert a client into a project, manage requirements/tasks, and allow a client to see only their permitted data.

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2. Objectives, Scope & Functional Definition

2.1 Learning Objectives

- Build a complete PERN application with React on the frontend, Node.js/Express on the backend and PostgreSQL as the relational database.
- Implement secure JWT authentication with access/refresh token handling and protected routes.
- Use Redux Toolkit for predictable global state, authentication state, cached entity data and UI state.
- Design reusable CRUD patterns for tables, forms, filters, pagination, modals, drawers and detail pages.
- Implement role-based access control at both frontend and backend levels.
- Design normalized relational data with foreign keys, indexes, status fields, timestamps and audit records.
- Create a responsive professional dashboard using Bootstrap 5 or Tailwind CSS plus Material Icons.

2.2 Core Scope

Area	Mandatory Capabilities
Authentication	Register, login, logout, forgot/reset password, email verification, JWT, refresh token, protected routes
Organization	Company, departments, employees, teams, roles, permissions
CRM	Leads, prospects, clients, contacts, companies, deals, sales pipeline
Projects	Projects, milestones, tasks, subtasks, assignments, deadlines, status
Client Portal	Client dashboard, project progress, tickets, messages, documents, invoices
Requirement Vault	Versioning, attachments, comments, screenshots/videos metadata, approvals
Operations	Notifications, activity log, search, filters, pagination, audit trail
Reporting	KPIs, pipeline chart, project status, task workload, revenue/invoice summary

2.3 Non-Functional Requirements

- Responsive at desktop, tablet and mobile breakpoints.
- Consistent loading, empty, success and error states.
- Form validation on the client and server; never trust frontend validation alone.
- API responses should use a consistent JSON shape and HTTP status codes.
- Passwords must be hashed; secrets must be stored in environment variables.
- Role checks must be enforced in backend middleware even if the UI hides unauthorized buttons.
- Database queries must use parameterized/ORM operations and proper foreign-key constraints.

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3. Technology Stack & System Architecture

Layer	Recommended Technology	Use
Frontend	React.js + Vite	SPA, routing, pages and reusable components
Styling	Bootstrap 5 OR Tailwind CSS	Responsive layout and utility styling
Icons	Material Icons / Material Symbols	Dashboard navigation, actions and status visuals
State	Redux Toolkit	Auth, CRM entities, UI state and async operations
API Client	Axios	HTTP requests, interceptors and token handling
Backend	Node.js + Express.js	REST API, middleware, controllers and services
Database	PostgreSQL	Relational persistence and reporting
ORM	Sequelize	Models, associations, migrations and queries
Auth	JWT + bcrypt	Access/refresh authentication and password hashing
Validation	Joi / Zod / express-validator	Server-side request validation
Uploads	Multer + storage provider	Requirement and document attachments
Testing	Postman + Jest/Supertest (recommended)	API and critical-flow verification
Version Control	Git + GitHub	Branches, commits, pull requests and history

3.1 High-Level Architecture

Client	API	Business Logic	Database
React Pages Components Redux Toolkit Axios ProtectedRoute	REST /api/v1 JWT middleware RBAC middleware Validation	Controllers Services Repositories/Models Audit service Notification service	PostgreSQL Users CRM Projects Requirements Invoices Tickets Audit logs

3.2 Request Flow

Browser -> React page -> Redux thunk -> Axios instance -> Express route -> Auth/RBAC middleware -> Controller -> Service -> Sequelize model -> PostgreSQL -> Service -> Controller -> JSON response -> Redux state -> UI.

3.3 Environment Variables

SERVER_PORT, DATABASE_URL, JWT_ACCESS_SECRET, JWT_REFRESH_SECRET, JWT_ACCESS_EXPIRES, JWT_REFRESH_EXPIRES, SMTP_HOST, SMTP_USER, SMTP_PASS, CLIENT_URL, STORAGE_BUCKET/UPLOAD settings and any optional 2FA/email configuration.

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4. Role-Based Organization & Permission Matrix

4.1 Recommended Roles

Role	Primary Responsibility	Dashboard Focus
Super Admin	Full system control	Global KPIs, users, permissions, audit logs, settings
Admin / Operations	Organization and daily operations	Teams, clients, projects, reports, activity
Sales / Business Developer	Leads and deals	Pipeline, leads, prospects, follow-ups, conversions
Project Manager	Delivery management	Projects, milestones, tasks, deadlines, client communication
Developer / Team Member	Execution	Assigned tasks, requirements, comments, project progress
Accounts	Commercial operations	Invoices, payments, quotations, revenue summaries
Support Agent	Client support	Tickets, priorities, SLA, messages, resolution
Client	External portal user	Own projects, requirements, tickets, invoices, documents

4.2 Permission Model

Use permissions rather than hard-coding every action to a role. Example permissions: users.view, users.create, users.update, leads.view, leads.create, leads.update, leads.delete, deals.manage, projects.view, projects.manage, tasks.assign, requirements.approve, invoices.manage, tickets.manage, reports.view, audit.view.

Module	Admin	Sales	PM	Developer	Accounts	Support	Client
Users/Roles	CRUD	-	-	-	-	-	-
Leads/Deals	CRUD	CRUD	R	-	-	-	-
Projects	CRUD	R	CRUD	R/U	R	R	R (own)
Tasks	CRUD	R	CRUD	R/U (assigned)	R	R	R (allowed)
Requirements	CRUD	R	CRUD	R/U	-	R	R/A (own)
Invoices	CRUD	R	R	R	CRUD	-	R (own)
Tickets	CRUD	R	R	R	R	CRUD	CRUD (own)
Reports/Audit	Full	Sales	Delivery	-	Finance	Support	Own

Rule: frontend visibility is for user experience; backend authorization is the real security boundary.

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5. Authentication, Authorization & Security

5.1 Authentication Screens

Screen	Component Name	Key Elements
Login	LoginPage / LoginForm	Email, password, remember option, validation, forgot password link
Register	RegisterPage / RegisterForm	Name, email, password, confirm password, terms
Verify Email	VerifyEmailPage	Token verification, resend email, status feedback
Forgot Password	ForgotPasswordPage	Email input, rate-limit feedback
Reset Password	ResetPasswordPage	Token, new password, confirmation, strength rules
2FA	TwoFactorPage	OTP/verification code, recovery path (advanced)
Logout	AuthService.logout	Clear state, revoke/rotate refresh token where applicable

5.2 Token Strategy

- Short-lived access token is used for API authorization.
- Longer-lived refresh token is used to obtain a new access token; store it securely according to the chosen deployment strategy.
- Axios response interceptor detects unauthorized access and attempts one controlled refresh before redirecting to login.
- Backend middleware verifies token signature, expiration and user status.
- Refresh-token rotation/revocation is recommended for a production-style implementation.

5.3 Security Controls

- bcrypt password hashing; never store plaintext passwords.
- RBAC middleware: authenticate -> authorize(permission) -> controller.
- Rate-limit login and password reset endpoints.
- Validate uploaded file type/size and never trust the filename.
- Use helmet, CORS configuration, secure cookies where appropriate, request logging and centralized error handling.
- Audit sensitive actions such as role changes, client approval, invoice changes and deletion.

5.4 Error Response Contract

Example: `{"success": false, "message": "Validation failed", "errors": [{"field": "email", "message": "Invalid email"}], "code": "VALIDATION_ERROR"}`

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6. CRM Core — Leads, Clients, Companies & Sales Pipeline

6.1 CRM Entities

Entity	Important Fields	Core Actions
Lead	name, email, phone, source, owner, score, status, notes	Create, assign, qualify, convert, archive
Prospect	company, contact, requirement summary, value, probability	Qualify, follow-up, convert to client
Company	name, industry, website, address, owner	CRUD, contacts, deals, projects
Contact	company/client, name, role, email, phone	CRUD, primary contact, communication
Client	company, billing info, account manager, status	CRUD, portal access, projects
Deal	title, value, stage, probability, expected close, owner	CRUD, stage movement, forecasting
Activity	type, subject, date, owner, related entity	Call, email, meeting, note, follow-up

6.2 Pipeline Stages

New Lead -> Contacted -> Qualified -> Proposal Sent -> Negotiation -> Won / Lost. Each stage change should record actor, timestamp and optional note.

6.3 CRM Screens

- LeadsPage: searchable table, filters, status chips, owner, source and bulk actions.
- LeadDetailsPage: profile, activity timeline, notes, attachments and convert action.
- PipelinePage: Kanban columns with drag/drop stage changes and deal cards.
- CompaniesPage / CompanyDetailsPage: contacts, deals, projects and activity.
- ClientsPage / ClientDetailsPage: portal status, projects, invoices, tickets and documents.
- ActivitiesPage: calls, meetings, follow-ups and overdue activity list.

6.4 Conversion Flow

Lead -> Qualification -> Prospect -> Deal -> Won -> Client record -> Client portal account (optional) -> Project creation -> Requirement Vault -> Delivery workflow.

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7. Project Management Module

7.1 Project Data

Object	Example Fields	Relationships
Project	code, name, clientId, PM, startDate, endDate, budget, status	Client, team, milestones, tasks, requirements
Milestone	projectId, title, dueDate, status, progress	Project, tasks
Task	projectId, milestoneId, title, assignee, priority, dueDate, status	Project, milestone, user, subtasks
Subtask	taskId, title, assignee, status	Task, user
Assignment	projectId/taskId, userId, role, allocation	User + work item
Project Activity	projectId, actor, action, timestamp	Project audit trail

7.2 Project Statuses

Planning -> Active -> On Hold -> In Review -> UAT -> Completed -> Archived. Status changes should be permission-controlled and logged.

7.3 Project Screens

- ProjectsPage: cards/table, filters by client, PM, status and deadline.
- ProjectDetailsPage: Overview, Team, Milestones, Tasks, Requirements, Documents, Invoices, Tickets, Activity.
- MilestonesPage: progress bars and deadline indicators.
- TaskBoardPage: Kanban by status; TaskDetailsDrawer for assignment and comments.
- MyWorkPage: developer-specific assigned tasks and due dates.
- ProjectReportsPage: completion %, overdue tasks, milestone health and team workload.

7.4 Task Rules

- Only permitted users can assign tasks.
- A task may have one primary assignee and optional watchers.
- Due-date changes are logged.
- Completed tasks can require review before final closure.
- Comments and attachments remain associated with the task/project for traceability.

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8. Client Portal

8.1 Client Experience

The client receives a separate portal experience with a restricted navigation menu. The client must never receive internal employees, internal notes, unrelated clients or system-wide reports.

Client Screen	Component	Purpose
Client Dashboard	ClientDashboardPage	Project progress, open tickets, pending approvals, recent activity
My Projects	ClientProjectsPage	List of projects belonging to the client
Project View	ClientProjectDetailsPage	Milestones, tasks allowed for client view, progress and documents
Requirements	ClientRequirementsPage	Submit/view requirements and approval status
Invoices	ClientInvoicesPage	Invoice list, status and downloadable documents
Support	ClientTicketsPage	Create ticket, reply, priority and status
Messages	ClientMessagesPage	Project/client communication
Documents	ClientDocumentsPage	Approved/shared files only
Profile	ClientProfilePage	Contact information and password

8.2 Client Access Rule

Every client API query should be scoped to the authenticated clientId. Do not rely on a clientId sent from the browser. The backend derives the permitted client identity from the authenticated user/session.

8.3 Client Approval Flow

Requirement submitted -> PM reviews -> clarification requested OR accepted for implementation -> version is frozen -> client approval recorded -> development task created/updated -> implementation -> review/UAT -> completion.

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9. Requirement Vault — Signature Feature

9.1 Requirement Record

Field	Description
Requirement Code	Unique identifier, e.g. REQ-104
Title / Description	Business/technical requirement text
Project	Parent project
Submitted By	Client or internal user
Version	Incrementing version number
Status	Draft, Submitted, Clarification, Approved, Rejected, Implemented
Priority	Low, Medium, High, Critical
Attachments	Screenshots, PDFs, images, videos metadata, documents
Comments	Threaded discussion
Approval	Approver, date/time, decision and optional note
Change Log	What changed between versions

9.2 Example

Requirement #REQ-104 — Add Dark Mode

Version: 3 | Status: Approved | Priority: Medium

Client approval: Approved by authorized client user on 25 Aug 2026

Change history: V1 basic toggle -> V2 theme persistence -> V3 accessibility contrast and system preference.

9.3 Requirement Screens

- RequirementsPage — searchable/version-aware list.
- RequirementCreatePage — structured form and attachment upload.
- RequirementDetailsPage — description, versions, comments, attachments and approval panel.
- RequirementVersionDrawer — compare current version against previous version.
- ApprovalModal — approve/reject/request clarification with reason.
- RequirementActivityTimeline — audit-style chronological history.

9.4 Why It Is Unique

This module connects sales expectations to delivery execution. It reduces the common software-house problem where requirements are scattered across WhatsApp, email, screenshots and informal chats.

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10. Full Dashboard, Analytics & Navigation

10.1 Global Dashboard Layout

Recommended shell: **Sidebar + Topbar + Breadcrumb + Page Header + Main Content + Notification Drawer**. Sidebar items should change by role. Topbar should contain global search, notifications, profile menu and quick-create button.

Dashboard Widget	Data Source	Behavior
Total Leads	leads	Count + trend
Pipeline Value	deals	Stage/value aggregation
Active Projects	projects	Count by status
Tasks Due	tasks	Due today / overdue
Revenue / Invoices	invoices	Monthly summary
Project Health	projects + tasks	Healthy / at-risk / overdue
Lead Conversion	leads + deals	Conversion percentage
Recent Activity	activities/audit	Timeline
Team Workload	tasks/assignments	Per-user open workload
Support Tickets	tickets	Open, pending, urgent

10.2 Dashboard Screen Inventory

- DashboardLayout — application shell and responsive sidebar.
- Sidebar — role-aware navigation and active route state.
- Topbar — search, notifications, profile, quick actions.
- KpiCard — metric, icon, label, trend and link.
- PipelineChart — deals grouped by stage.
- ProjectHealthCard — progress and deadline health.
- TaskOverview — due/overdue/assigned filters.
- ActivityTimeline — recent actions.
- QuickCreateMenu — Lead, Client, Deal, Project, Task, Requirement, Ticket.
- NotificationDrawer — approvals, assignments, deadlines and ticket updates.

10.3 Material Icon Theme

Use Material Icons consistently: dashboard, people, business, contact_page, work, task_alt, flag, description, receipt_long, support_agent, chat, folder, notifications, settings, security, analytics, calendar_month, search, add, edit, delete, visibility, more_vert and logout. Keep icon usage semantic and consistent across sidebar, buttons and cards.

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11. Frontend Screen & Component Inventory

11.1 Routing

Recommended route groups: /auth/*, /app/* for internal users, and /portal/* for clients. Every protected route should pass through authentication and permission checks.

Area	Page Components	Reusable Components
Auth	LoginPage, RegisterPage, VerifyEmailPage, ForgotPasswordPage, ResetPasswordPage, TwoFactorPage	AuthForm, PasswordField, AuthGuard
Dashboard	DashboardPage	KpiCard, ChartCard, ActivityTimeline, QuickCreate
CRM	LeadsPage, LeadDetailsPage, CompaniesPage, CompanyDetailsPage, ContactsPage, ClientsPage, ClientDetailsPage, PipelinePage	DataTable, SearchBar, FilterBar, StatusBadge, DealCard
Projects	ProjectsPage, ProjectDetailsPage, MilestonesPage, TaskBoardPage, MyWorkPage	ProjectCard, ProgressBar, TaskCard, TaskDrawer, AssigneeSelect
Requirements	RequirementsPage, RequirementCreatePage, RequirementDetailsPage	VersionBadge, ApprovalModal, AttachmentList, CommentThread
Finance	InvoicesPage, InvoiceDetailsPage, QuotationsPage	InvoiceTable, PaymentStatus, CurrencyField
Support	TicketsPage, TicketDetailsPage	PriorityBadge, TicketThread, SLAIndicator
Admin	UsersPage, RolesPage, PermissionsPage, DepartmentsPage, SettingsPage, AuditLogPage	RoleMatrix, PermissionCheckbox, AuditTable
Client Portal	ClientDashboardPage, ClientProjectsPage, ClientProjectDetailsPage, ClientRequirementsPage, ClientInvoicesPage, ClientTicketsPage, ClientDocumentsPage	PortalHeader, ClientProjectCard, ApprovalPanel

11.2 UI Standards

- Use consistent spacing, typography, button hierarchy and status colors.
- Tables need loading skeleton, empty state, pagination, sorting and filters.
- Forms need inline validation and disabled submit while saving.
- Destructive actions require confirmation and should show a clear success/error result.
- Detail pages should use tabs when the entity has multiple related datasets.
- Mobile layout must collapse sidebar and convert wide tables into responsive views where practical.

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12. Redux Toolkit Store, Slices & API State

12.1 Store Structure

Suggested global store:

```
store/
index.js
slices/authSlice.js
slices/userSlice.js
slices/leadSlice.js
slices/companySlice.js
slices/clientSlice.js
slices/dealSlice.js
slices/projectSlice.js
slices/taskSlice.js
slices/requirementSlice.js
slices/invoiceSlice.js
slices/ticketSlice.js
slices/notificationSlice.js
slices/uiSlice.js
```

12.2 Slice Responsibilities

Slice	State Examples	Async Operations
authSlice	user, accessToken, authenticated, loading	login, register, verify, refresh, logout
leadSlice	items, selected, filters, pagination, loading	list, get, create, update, delete, convert
projectSlice	projects, currentProject, filters	CRUD, status change, assign team
taskSlice	tasks, board, filters	CRUD, assign, move status, comments
requirementSlice	requirements, versions, selected	CRUD, upload, approve, version create
notificationSlice	items, unreadCount	list, markRead, markAllRead
uiSlice	sidebarOpen, modal, theme, globalLoader	local UI state

12.3 Redux Rules

- Do not duplicate server data in many unrelated slices without a reason.
- Use createAsyncThunk or RTK Query; if RTK Query is selected, document the decision and use it consistently.
- Keep filters/pagination close to the feature that owns them.
- Reset feature state on logout where necessary.
- Use selectors for derived data such as dashboard counts and permissions.

12.4 API Call Flow

Component -> dispatch(fetchLeads(params)) -> thunk -> api/leadApi.js -> Axios -> endpoint -> response -> fulfilled/rejected -> reducer -> selector -> component render.

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13. REST API Structure & Endpoint Plan

13.1 Base Structure

/api/v1/auth | /api/v1/users | /api/v1/roles | /api/v1/permissions | /api/v1/leads | /api/v1/companies | /api/v1/contacts | /api/v1/clients | /api/v1/deals | /api/v1/projects | /api/v1/milestones | /api/v1/tasks | /api/v1/requirements | /api/v1/invoices | /api/v1/tickets | /api/v1/messages | /api/v1/documents | /api/v1/notifications | /api/v1/reports | /api/v1/audit-logs

Method	Endpoint	Purpose
POST	/auth/register	Create user account
POST	/auth/login	Authenticate and issue tokens
POST	/auth/refresh	Issue a new access token
POST	/auth/forgot-password	Create reset workflow
POST	/auth/reset-password	Set new password
GET	/leads	List leads with search/filter/pagination
POST	/leads	Create lead
PATCH	/leads/:id	Update lead
POST	/leads/:id/convert	Convert lead to prospect/client workflow
GET	/deals/pipeline	Return pipeline grouped by stage
POST	/projects	Create project
GET	/projects/:id	Project details with permitted relations
POST	/tasks	Create task
PATCH	/tasks/:id/status	Change task status
POST	/requirements	Create requirement
POST	/requirements/:id/versions	Create new requirement version
POST	/requirements/:id/approve	Approve/reject/request clarification
GET	/dashboard/summary	Role-scoped KPI data
GET	/audit-logs	Authorized audit log search

13.2 Backend Middleware Order

request -> CORS -> helmet/security -> JSON parser -> request logger -> auth middleware (protected routes) -> permission middleware -> validation middleware -> route controller -> centralized error handler.

13.3 Standard API Response

Success: {"success": true, "message": "Lead created", "data": {...}}
List: {"success": true, "data": [...], "meta": {"page": 1, "limit": 10, "total": 120}}

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14. Backend Architecture, Folder Structure & Database

14.1 Express Project Structure

```
server/
src/
config/ (database, env)
controllers/ (auth, lead, client, project...)
services/ (business logic)
models/ (Sequelize models)
routes/ (versioned route modules)
middleware/ (auth, rbac, validation, upload, errorHandler)
validators/
utils/ (tokens, email, response, pagination)
uploads/ (temporary/local development)
app.js
server.js
migrations/
seeders/
```

14.2 Database Core Tables

Table	Key Columns / Relationships
users	id, name, email, passwordHash, status, roleId, departmentId, createdAt
roles	id, name, description
permissions	id, key, description
role_permissions	roleId, permissionId
companies	id, name, industry, website, address, ownerId
contacts	id, companyId/clientId, name, role, email, phone
leads	id, companyId, contactId, ownerId, source, status, score
deals	id, leadId/companyId, ownerId, stage, value, probability
clients	id, companyId, accountManagerId, portalUserId, status
projects	id, clientId, pmlId, name, code, dates, budget, status
milestones	id, projectId, title, dueDate, status, progress
tasks	id, projectId, milestoneId, assigneeId, title, status, priority, dueDate
requirements	id, projectId, submittedBy, title, currentVersion, status
requirement_versions	id, requirementId, versionNo, body, createdBy
requirement_approvals	id, requirementId/versionId, approverId, decision
invoices	id, clientId, projectId, number, amount, status, dueDate
tickets	id, clientId, projectId, createdBy, priority, status
messages	id, projectId/ticketId, senderId, body, createdAt
notifications	id, userId, type, title, readAt
audit_logs	id, actorId, action, entityType, entityId, metadata, createdAt

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15. Sequelize Associations & CRUD Standards

15.1 Key Associations

- Role hasMany Users; User belongsTo Role.
- Company hasMany Contacts, Leads, Deals and Clients.
- Client hasMany Projects, Invoices and Tickets.
- Project belongsTo Client and PM; Project hasMany Milestones, Tasks, Requirements, Documents and Tickets.
- Milestone belongsTo Project and hasMany Tasks.
- Task belongsTo Project, Milestone and User(assignee); Task hasMany Subtasks and Comments.
- Requirement hasMany RequirementVersions, Attachments, Comments and Approvals.
- User hasMany AuditLogs and Notifications.

15.2 CRUD Pattern

Create: validate -> authorize -> normalize -> create -> audit -> notify -> return resource.

Read: authorize -> apply scope -> filter/search -> paginate -> include only permitted relations.

Update: validate -> authorize -> compare old/new values -> update -> audit -> notify if important.

Delete: authorize -> check dependencies -> soft-delete/archive where appropriate -> audit.

15.3 Pagination & Filtering

Standard query parameters: ?page=1&limit=10&search=acme&status=active&sortBy=createdAt&sortOrder=desc. Return total, page, limit and totalPages.

15.4 Soft Delete

For business records such as clients, projects and requirements, prefer archive/soft-delete semantics rather than destructive deletion. Preserve historical records and audit trails.

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16. End-to-End Business Workflows

Workflow	Flow
Lead to Client	Lead created -> assigned to Sales -> follow-up -> qualified -> prospect -> deal -> proposal -> negotiation -> won -> client created -> portal invited
Client to Project	Client selected -> project created -> PM assigned -> team assigned -> milestones -> requirements -> tasks -> delivery
Requirement	Client/internal submit -> PM review -> clarification OR approved -> version frozen -> task(s) linked -> implementation -> review -> completed
Task	Create -> assign -> developer starts -> progress/comments -> review -> done -> activity logged
Invoice	Draft -> issue -> sent -> partially/fully paid OR overdue -> finance report
Support	Client opens ticket -> support assigns -> response -> waiting client / in progress -> resolved -> closed
Notification	Event occurs -> notification created -> user sees unread badge -> opens -> mark read -> linked detail page

16.1 Example: REQ-104 Flow

Client submits "Add Dark Mode" -> requirement created as V1 -> PM asks for contrast requirements -> client updates V2 -> client submits screenshots -> V3 is reviewed -> authorized client approves -> PM creates task "Implement Theme Provider" -> developer implements -> reviewer validates -> requirement marked Implemented.

16.2 Dashboard Data Flow

Login identifies role -> dashboard endpoint returns only role-permitted aggregates -> KPI cards render -> user opens a KPI -> application navigates to filtered detail list -> user updates record -> Redux refreshes affected data -> audit event is created.

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17. Reports, Notifications, Documents & Audit

17.1 Reports

Report	Filters / KPIs
Sales Report	Lead source, owner, stage, conversion, pipeline value
Project Report	Client, PM, status, progress, overdue milestones/tasks
Team Workload	User, open tasks, overdue tasks, completion rate
Finance Summary	Invoice status, billed, paid, outstanding, overdue
Support Report	Open tickets, priority, response/resolution status
Requirement Report	Approved, pending, rejected, clarification, version count
Activity Report	User, action, entity, date range

17.2 Notifications

- New lead assigned.
- Deal stage changed.
- Project/team assignment.
- Task assigned or deadline approaching.
- Requirement submitted/approved/rejected.
- Invoice issued or overdue.
- Ticket assigned or replied.
- Client approval requested.

17.3 Document Management

Store metadata in PostgreSQL and files in a suitable storage layer. Record original filename, MIME type, size, uploader, entity type, entity id and access visibility. Client-facing documents must be explicitly marked as shared/approved.

17.4 Audit Log

Capture actor, action, entity type, entity id, timestamp, IP/user-agent where appropriate, and a safe before/after summary. Do not store secrets or raw passwords in logs.

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18. Testing, Validation & Acceptance Criteria

18.1 Mandatory Test Scenarios

Test	Expected Result
Register with invalid email	Validation error; no user created
Login with wrong password	401 response; generic authentication message
Access admin endpoint as client	403 forbidden
Open another client's project by URL id	404/403 according to policy; no data leakage
Create lead without required fields	400 validation response
Convert qualified lead	Client/deal relationship created correctly
Move deal stage	Stage updated and activity logged
Assign task without permission	403; assignment unchanged
Approve requirement	Approval record created and requirement status changes
Client views invoices	Only that client's invoices returned
Expired access token	Refresh flow or login redirect
Upload invalid file	Rejected by type/size validation
Delete/Archive record	Permission check + audit entry

18.2 UI Acceptance

- No broken routes, console errors or missing key states.
- Every CRUD form has loading and error handling.
- Tables support search/filter/pagination where required.
- Unauthorized navigation is hidden and direct URL access is blocked.
- Dashboard cards link to the relevant detail lists.
- All core pages work at desktop and mobile widths.

18.3 Definition of Done

A feature is done when database model/migration, API route, validation, authorization, frontend page, Redux/API state, loading/error/empty states, responsive UI, audit behavior and at least one tested happy path are implemented.

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19. Implementation Timeline — 25 August to 02 September 2026

Date	Focus	Expected Output
25 Aug	Planning + setup	Repo, React/Vite, Express server, PostgreSQL, Sequelize, env, base layout, Git workflow
26 Aug	Authentication + RBAC	Register/login/logout, JWT, refresh, protected routes, roles/permissions
27 Aug	Organization + CRM	Users, companies, contacts, leads, clients, deals, pipeline
28 Aug	Projects + tasks	Projects, milestones, tasks, assignments, Kanban and project details
29 Aug	Requirement Vault + client portal	Requirement versions/approvals, portal dashboard, client projects
30 Aug	Finance + support + notifications	Invoices, tickets, messages, notifications, documents
31 Aug	Dashboard + reports	KPIs, charts, activity, workload, reports and global search
01 Sep	Testing + polish	API tests, validation, RBAC testing, responsive fixes, error states
02 Sep	Final integration	Production-style demo, README, seed data, screenshots, final build and submission package

19.1 Daily Development Rule

Each day should end with: working code pushed to Git, short progress note, unresolved issues listed, and the next day's priorities identified. Avoid building every screen first and connecting APIs at the end.

19.2 Suggested Git Branches

```
main -> stable releases | develop -> integration | feature/auth | feature/crm | feature/projects | feature/requirements
| feature/client-portal | feature/dashboard | fix/*
```

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20. Final Deliverables & Submission Package

20.1 Required Deliverables

- Frontend source code (React/Vite) with documented folder structure.
- Backend source code (Node.js/Express) with routes, controllers, services and middleware.
- PostgreSQL database schema with Sequelize models/migrations and seed data.
- Authentication and role-based authorization implementation.
- Responsive dashboard and all mandatory core modules.
- Requirement Vault with versioning and approval flow.
- Client portal with scoped access.
- API collection (Postman or equivalent) covering major endpoints.
- README containing installation, environment variables, database setup, seed accounts and run commands.
- Screenshots or short demo video of major flows.
- Final Git repository with clean commit history.

20.2 Suggested Seed Accounts

Account	Role	Purpose
admin@demo.local	Super Admin	Full system demonstration
sales@demo.local	Sales	Lead/deal pipeline demonstration
pm@demo.local	Project Manager	Project/task/requirement demonstration
developer@demo.local	Developer	Assigned work demonstration
accounts@demo.local	Accounts	Invoice/report demonstration
support@demo.local	Support	Ticket demonstration
client@demo.local	Client	Portal demonstration

Use clearly fake/demo credentials in the project documentation; never commit real company passwords, SMTP credentials or production secrets.

20.3 Demo Sequence

Login as Admin -> dashboard -> create lead -> qualify -> convert -> create client -> create project -> add requirement -> approve -> create/assign task -> switch to Developer -> complete task -> switch to Client -> review project/requirement -> open support ticket -> switch to Support -> resolve ticket -> return to Admin -> show audit/report.

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21. Advanced Add-ons & Extra Credit

Add-on	Description	Priority
2FA	OTP/app-based second factor for privileged users	Advanced
Email Automation	Lead follow-up, invoice reminders, approval emails	Advanced
Global Search	Search across leads, clients, projects, requirements and tickets	High
Calendar	Meetings, follow-ups, deadlines and milestone dates	Medium
Time Tracking	Developer work logs against tasks/projects	Medium
Quotations	Proposal/quotation builder linked to deals	High
Recurring Invoices	Subscription/maintenance billing support	Advanced
AI Lead Summary	Summarize notes and suggest next action	Optional
AI Requirement Summary	Summarize long requirement history	Optional
Webhooks	Notify external systems about deal/project events	Advanced
Dark Mode	Persistent UI theme with accessibility contrast	Medium
Export	CSV/PDF exports for reports and tables	High
Activity Mentions	@user comments and notifications	Medium
Kanban Drag & Drop	Deal/task movement with optimistic UI and rollback	High

Recommended Product Enhancements

- Global command palette (Ctrl/Cmd + K) for fast navigation.
- Saved filters for Sales and Project Managers.
- Client approval certificate/history per requirement version.
- At-risk project indicator based on overdue milestones and task completion.
- Lead scoring using configurable rules.
- Reusable document templates for quotations, invoices and project kickoff.

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22. Final Architecture Checklist

Layer	Checklist
Frontend	React + Vite; routes; protected routes; role-aware navigation; reusable components; responsive layout; Material Icons
State	Redux Toolkit; auth state; feature slices; loading/error states; selectors; logout reset
API	Axios instance; auth interceptor; REST endpoints; validation; pagination; consistent response contract
Backend	Express; routes; controllers; services; middleware; centralized errors; RBAC; logging
Database	PostgreSQL; Sequelize; migrations; associations; indexes; seeders; constraints
Security	JWT; refresh flow; bcrypt; RBAC; rate limits; secure config; upload validation; audit logs
CRM	Leads; prospects; companies; contacts; clients; deals; pipeline; activities
Delivery	Projects; milestones; tasks; assignments; deadlines; project status
Client	Portal; scoped data; requirements; invoices; tickets; documents; messages
Signature Feature	Requirement Vault; versions; attachments; comments; approvals; change history
Dashboard	KPIs; charts; activity; workload; project health; notifications; quick actions
Quality	Testing; responsive design; empty/loading/error states; README; seed data; demo flow

Final quality bar: the project should look and behave like an internal software-house platform, not a tutorial CRUD app. The strongest submissions will demonstrate clean architecture, secure authorization, realistic relational data, thoughtful UX and a complete end-to-end workflow.

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23. Official Project Brief — Approval & Issue

This document defines the final project scope for the **DevFlow CRM — CRM Built Specifically for Software Houses** Full-Stack Development Internship project at U Devs for the 2026 program.

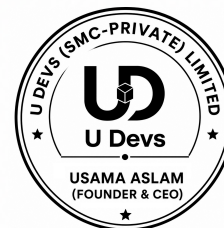
The project window is **25 August 2026 to 02 September 2026**. The implementation is expected to follow the PERN architecture and the functional, security, role-based, dashboard, CRM, project-management, client-portal and Requirement Vault requirements described in this specification.

Official Project Details	Value
Organization	U Devs SMC PVT Limited
Project	DevFlow CRM
Program	Full-Stack Development Internship — Final Project
Technology	PERN + JWT + Redux Toolkit + Bootstrap/Tailwind + Material Icons
Start	25 August 2026
End	02 September 2026
Website	www.udevs.company
Secondary Website	www.udevs-hub.com
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Usama Aslam
Founder & CEO — U Devs

U Devs SMC PVT Limited

Document status: **Official Final Project Specification / Assignment Brief**

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